

Consider the hydrogen halides HF, HCl, HBr and HI. Which of the statements about them is true?

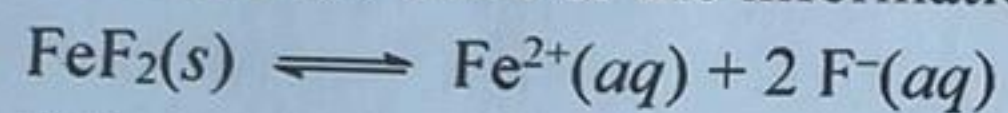
- A) They are all strong acids
- B) They are all weak acids
- C) The boiling point increases with molar mass.
- D) The bond energy increases with molar mass.
- E) none of above

A22. In the gas phase reaction below, NH_3 is acting as a(n) _____ base but not as a(n) _____ base.

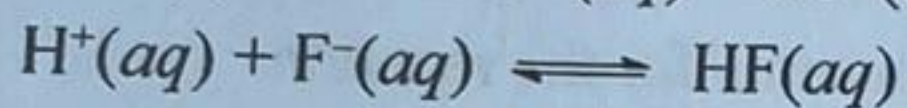


- A) Arrhenius, Brønsted-Lowry
- B) Lewis, Brønsted-Lowry
- C) Brønsted-Lowry, Lewis
- D) Arrhenius, Lewis
- E) Lewis, Arrhenius

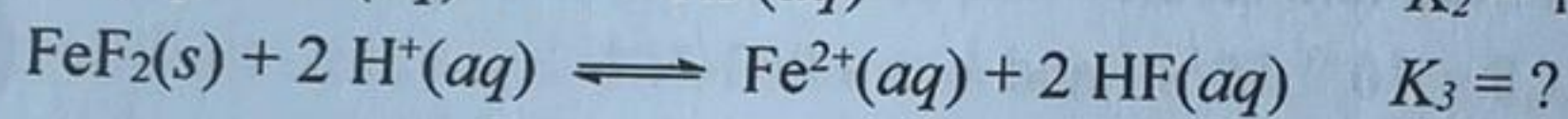
A23. On the basis of the information below, the dissolution of $\text{FeF}_2(s)$ in acidic solution is _____.



$$K_1 = 2 \times 10^{-6}$$



$$K_2 = 1 \times 10^3$$



$$K_3 = ?$$

- A) thermodynamically favorable, because $K_2 > 1$
- B) thermodynamically favorable, because $K_3 > 1$
- C) not thermodynamically favorable, because $K_1 < 1$
- D) not thermodynamically favorable, because $K_3 < 1$
- E) Insufficient data are given.

A24. Is it possible for a solution to have a negative pH? If so, would that pH signify a basic or acidic solution?

A) No, the pH range is 1~14

B) No, the definition of pH does not permit it to have a negative value.

C) Yes, for any solution with a concentration of strong base greater than 1 M.

D) Yes, for any solution with a concentration of strong acid greater than 1 M.

E) Yes, for any solution with a concentration of strong acid smaller than 1 M.

A25. The equilibrium constant for a reaction is greater than 1.0 at temperatures above 500 K but less than 1.0 at temperatures below 500 K. What can be concluded about the values of ΔH° and ΔS° for the reaction? (Assume that ΔH° and ΔS° are independent of temperature.)

- A) $\Delta H^\circ > 0$ and $\Delta S^\circ > 0$
- B) $\Delta H^\circ > 0$ and $\Delta S^\circ < 0$
- C) $\Delta H^\circ < 0$ and $\Delta S^\circ > 0$
- D) $\Delta H^\circ < 0$ and $\Delta S^\circ < 0$
- E) $\Delta H^\circ > 0$ and $\Delta S^\circ = 0$

A15. A solution is prepared by dissolving 23.7 g of CaCl_2 in 375 g of water. The density of the resulting solution is 1.05 g/mL. The concentration of this solution is _____ m.

- A) 0.562 B) 5.70 C) 0.569 D) 63.2 E) 1.76

A16. One difference between first- and second-order reactions is that _____.

- _____ the half-life of a first-order reaction depends on $[A]_0$; the half-life of a second-order reaction does not depend on $[A]_0$
 _____ the rate of a first-order reaction depends on reactant concentrations; the rate of a second-order reaction does not depend on reactant concentrations
 _____ a first-order reaction can be catalyzed; a second-order reaction cannot be catalyzed
 _____ the half-life of a first-order reaction does not depend on $[A]_0$; the half-life of a second-order reaction depends on $[A]_0$
 E) None of the above are true.

A17. For the reaction $2 \text{A}(g) + \text{B}(g) \rightleftharpoons \text{C}(g)$, $K_c = 1.0 \times 10^{-20}$.

In an experiment, 1.0 mol each of A, B and C are placed in an empty 1.0 L container and then the container is quickly sealed. When equilibrium is established, which of the following will be true?

- A) $[A] < [B] < [C]$ B) $[A] > [B] > [C]$
 C) $[A] = [B] = [C]$ D) $[A] = [B] < [C]$ E) $[A] > [B] = [C]$

A solution of NaF is added dropwise to a solution that is 0.0111 M in Ba^{2+} . When the concentration of F^- exceeds _____ M, BaF_2 will precipitate. Neglect volume changes. For BaF_2 , $K_{sp} = 1.7 \times 10^{-6}$.

- A) 7.7×10^{-5} B) 1.5×10^{-4} C) 3.1×10^{-3} D) 1.2×10^{-2} E) 1.9×10^{-8}

A19. Consider the reaction: $\text{FeO}(s) + \text{Fe}(s) + \text{O}_2(g) \rightarrow \text{Fe}_2\text{O}_3(s)$, given the following table of thermodynamic data, determine the temperature (in $^\circ\text{C}$) above which the reaction is non-spontaneous.

Substance	ΔH_f° (KJ/mol)	S° (J/mol·K)
$\text{FeO}(s)$	-271.9	60.75
$\text{Fe}(s)$	0	27.15
$\text{O}_2(g)$	0	205.0
$\text{Fe}_2\text{O}_3(s)$	-822.16	89.96

- A) 756.3 B) 1235 C) 618.1 D) 2438
 E) The reaction is spontaneous at all temperatures.

The value of K_w at 37°C is 2.4×10^{-14} . Which of the following statements about the pH and pOH of pure water at 37°C is correct?

- A) $\text{pH} < 7$ and $\text{pOH} > 7$ because water dissociates to a lesser extent at 37°C than at 25°C .
 B) $\text{pH} > 7$ and $\text{pOH} < 7$ because the dissociation of water is endothermic and $\text{pH} + \text{pOH} = 14$.
 C) $\text{pH} = \text{pOH} < 7$ because water dissociates to a greater extent at 37°C than at 25°C .
 D) $\text{pH} = \text{pOH} = 7$ because $[\text{H}_3\text{O}^+] = [\text{OH}^-]$ for pure water.
 E) $\text{pH} = \text{pOH} > 7$ because water dissociates to a greater extent at 37°C than at 25°C .

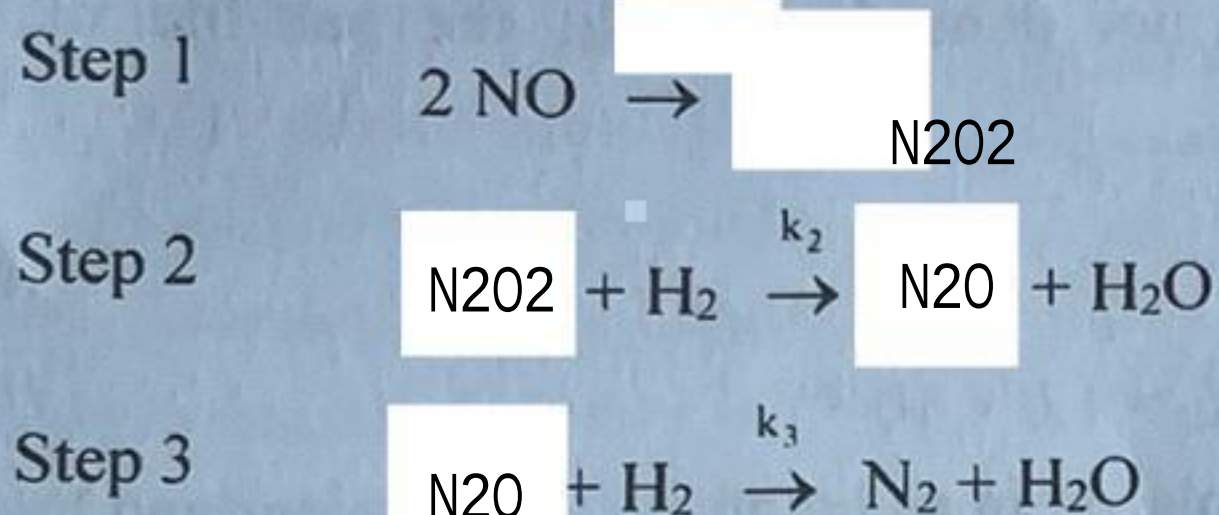
A9. The vapor pressure of pure water at 25 °C is 3.173 kPa. What is the vapor pressure (kPa) of water above a solution prepared by dissolving 18.0 g of glucose ($C_6H_{12}O_6$) in 95.0 g of water?

- A) 3.24 B) 3.11 C) 0.060 D) 0.059 E) 3.17

A10. What is the pH of a solution at 25 °C in which 35.0 mL of 0.067 M of $Ba(OH)_2$ is mixed with 50.0 mL of 0.050 M HBr?

- A) 1.59 B) 1.89 C) 7.00 D) 8.00 E) 12.41

A11. A possible mechanism for the gas phase reaction of NO and H_2 is as follows:



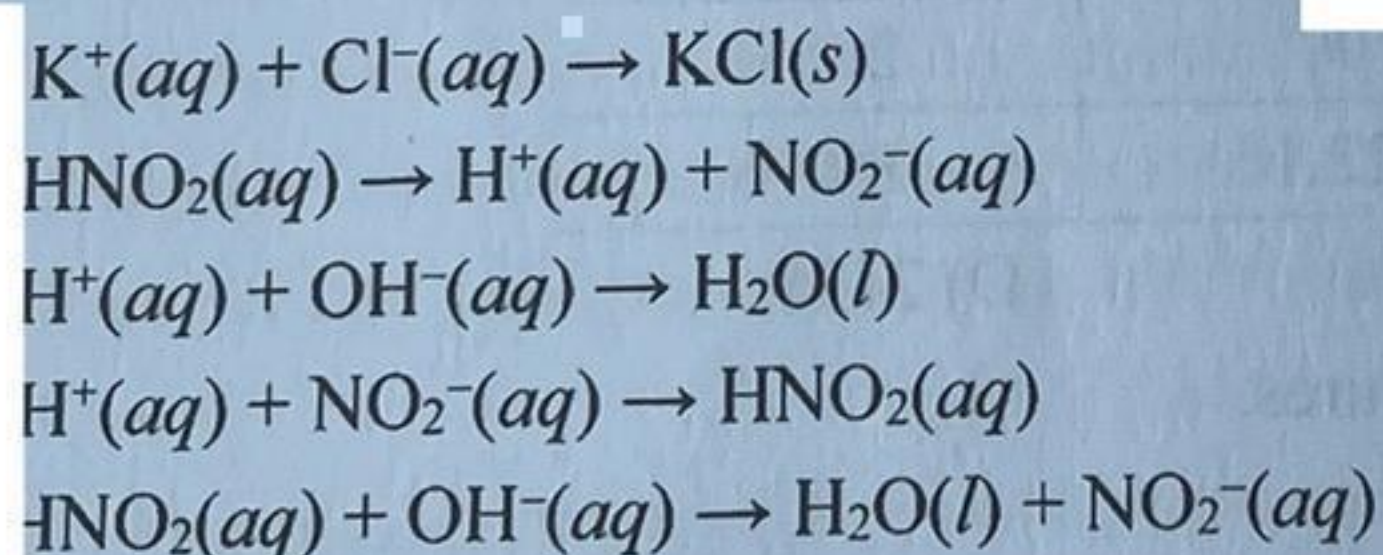
Which of the following statements concerning this mechanism is not directly supported by the information provided?

- A) Step 1 is the rate determining step.
 B) N_2O_2 is an intermediate.
 C) There is no catalyst in this reaction.
 D) The rate law for step 1 is $rate = k_1[NO]^2$
 E) All steps are bimolecular reactions.

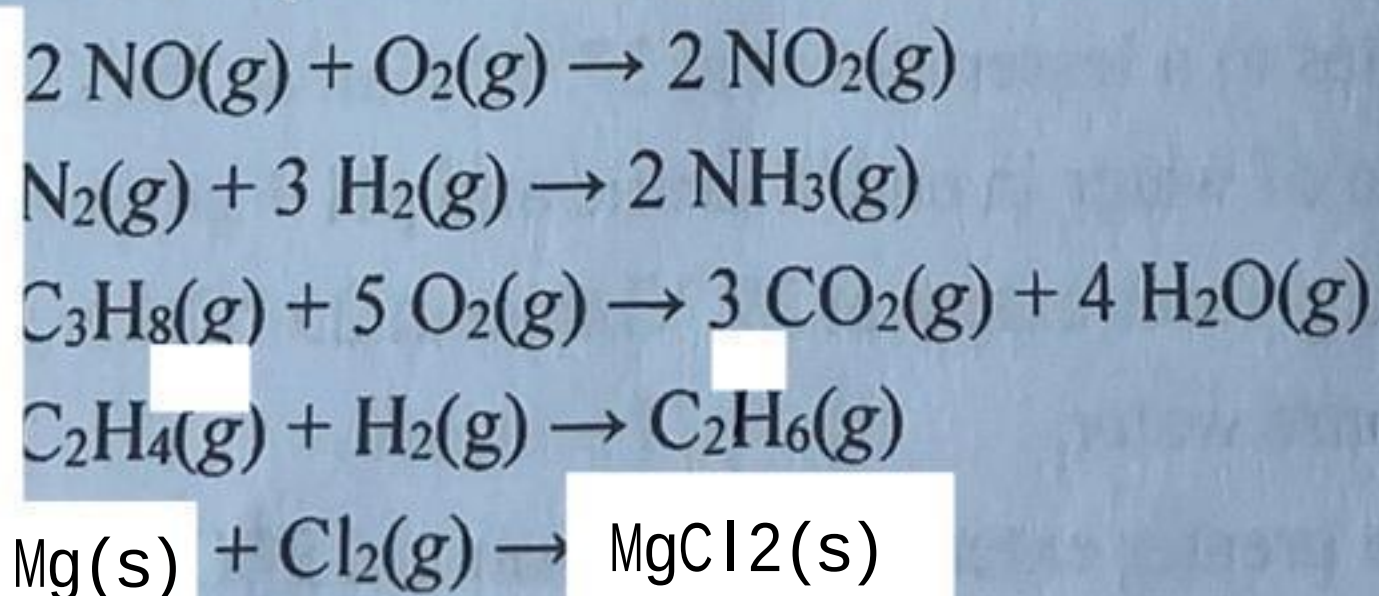
A12. The heat of fusion of H_2O is 6.01 kJ/mol. The heat capacity of water is 75.3 J/mol·K. The conversion of 50.0 g of ice at 0.00 °C to water at 15.0 °C requires _____ kJ of heat.

- A) 16.7 B) 3154 C) 3.14 D) 19.8 E) Insufficient data are given.

A13. A student prepares a solution by combining 100 mL of 0.30 M $HNO_2(aq)$ and 100 mL of 0.30 M $KNO_2(aq)$. Which of the following equations represents the reaction that best helps to explain why adding a few drops of 1.0 M $HCl(aq)$ does not significantly change the pH of the solution?



A14. ΔS is positive for the reaction _____.



A3. What is the magnitude of k for a first-order reaction at $60.00\text{ }^{\circ}\text{C}$ if $E_a = 121\text{ kJ/mol}$ and rate constant is $1.75 \times 10^{-1}\text{ s}^{-1}$ at $20.00\text{ }^{\circ}\text{C}$? ($R = 8.314\text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$)

- A) 385 B) 136 C) 68 D) 34 E) 17

A4. Which statements about viscosity is/are true?

Viscosity is the energy required to expand the surface area of a liquid by a unit amount of area.

Viscosity increases as temperature decreases.

Viscosity increases as intermolecular forces increase.

- A) (i) only B) (ii) and (iii) C) (i) and (iii) D) none E) all

A5. The data in the table below were obtained for the reaction: $A + B \rightarrow P$

Experiment	[A] (M)	[B] (M)	Initial Rate (M/s)
1	0.273	0.763	2.83
2	0.273	1.526	2.83
3	0.819	0.763	25.47

The rate law for this reaction is rate = _____.

- A) $k[A]^2[B]$ B) $k[A][B]$ C) $k[A]^2$ D) $k[A]^2[B]^2$ E) $k[P]$

A6. Based on molecular mass and dipole moment of the five compounds in the table below, which should have the highest boiling point?

Substance	Molecular Mass (amu)	Dipole Moment (D)
Propane, $\text{CH}_3\text{CH}_2\text{CH}_3$	44	0.1
Dimethylether, CH_3OCH_3	46	1.3
Methylchloride, CH_3Cl	50	1.9
Acetaldehyde, CH_3CHO	44	2.7
Acetonitrile, CH_3CN	41	3.9

- A) $\text{CH}_3\text{CH}_2\text{CH}_3$ B) CH_3OCH_3 C) CH_3CHO D) CH_3Cl E) CH_3CN

Let HA represent a weak monoprotic acid with $K_a = 1.0 \times 10^{-5}$. In an experiment, a 50.0 mL sample of 0.10 M HA(aq) is titrated with 0.10 M NaOH(aq). At which point during the titration are the equilibrium concentrations of H^+ and OH^- equal?

- A) after the addition of exactly 25.0 mL of NaOH(aq)
 B) after the addition of slightly less than 50.0 mL of NaOH(aq)
 C) after the addition of exactly 50.0 mL of NaOH(aq)
 D) after the addition of more than 50.0 mL of NaOH(aq)
 E) The equilibrium concentrations of H^+ and OH^- are never equal.

A8. The normal boiling point of water is $100.0\text{ }^{\circ}\text{C}$ and its molar enthalpy of vaporization is 40.67 kJ/mol . What is the change in entropy in the system in J/K when 39.3 grams of steam at 1 atm condenses to a liquid at the normal boiling point?

- A) -40.7 B) 88.8 C) 373 D) 238 E) 238

Course Name: Chemistry: the Central Science (CH105) Dept.: ChemistryExam Duration: 2 h Exam Paper Setter: _____

Question No.	A	B	C
Score	50	24	26

This exam paper contains 3 parts (Part A: 25 questions, Part B: 6 questions and Part C: 3 questions) and the score is 100 in total. (Please hand in your exam paper, answer sheet, and your scratch paper to the proctor when the exam ends.)

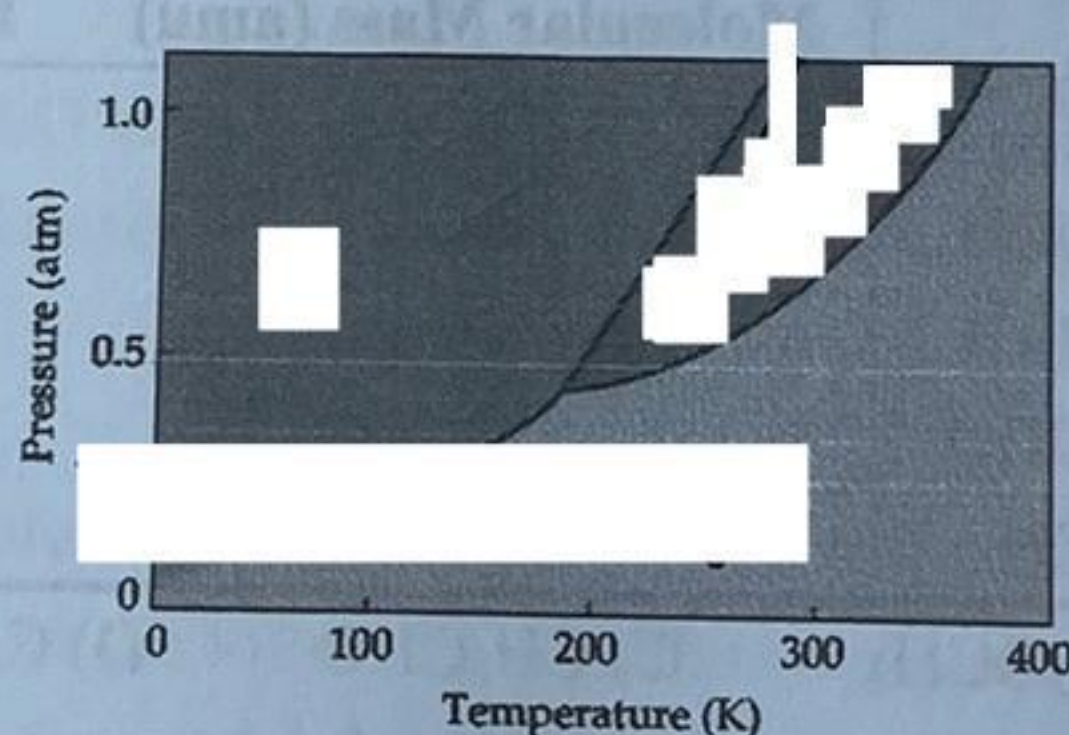
This test is closed-book, closed-notes, and closed-neighbors. A periodic table is provided. Note: please write down your answers in English only on the answer sheet.

Part A: Multiple Choice – Select the one that is best in each case. (50 Pts)

Number of questions: 25

Points per question: 2

A1. Based on the phase diagram of a hypothetical substance below, what happens to it as it is heated from 50 K to 200 K at 0.25 atm?



- A) It melts at about 125 K.
B) It sublimates at about 125 K.
C) It condenses at about 125 K.
D) It boils at about 125 K.
E) It reaches the triple point at about 200 K.

Which aqueous solution will have the lowest freezing point?

- A) 0.20 *m* sucrose ($C_{12}H_{22}O_{11}$)
B) 0.050 *m* $CaCl_2$
C) 0.10 *m* HCl
D) 0.15 *m* acetic acid (CH_3COOH)
E) 0.15 *m* $NaCl$

Part C: Long Answer – Write legibly and show work for all steps in the following problems. For mathematical relationships, show both the equations used and substitution of values into the equation. Maintain correct units throughout your calculations and report your answers with the correct significant figures. (26 Pts)

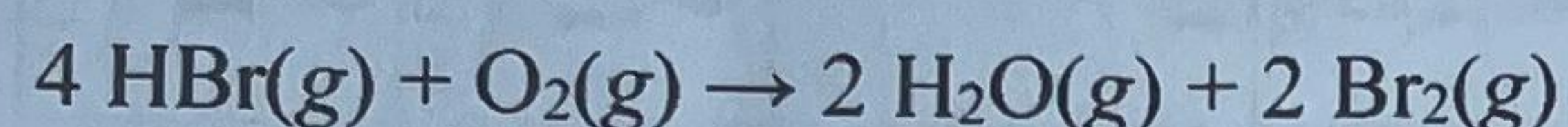
Number of questions: 3

C1: 9 Pts ; C2: 8 Pts ; C3: 9 Pts

C1. A 0.10 M solution of formic acid (HCOOH) has pH = 2.38 at 25 °C.

- (a) Calculate K_a for formic acid at 25 °C.
 (b) What is the percent ionization of formic acid at 25 °C?
 (c) Calculate the pH of a 0.20 M aqueous solution of HCOONa at 25 °C.

C2. Consider the gas-phase oxidation of HBr by O₂:

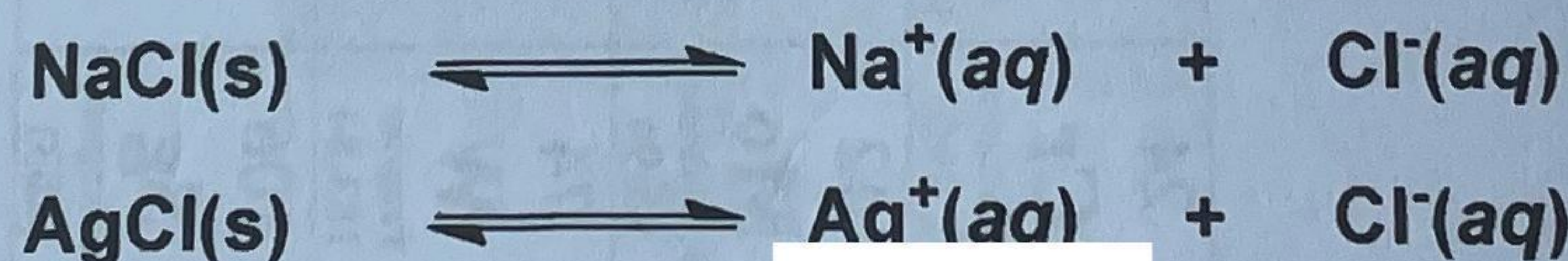


It is found that the reaction to be first order with respect to HBr and first order with respect to O₂. The following mechanism is proposed:

- i. $\text{HBr}(g) + \text{O}_2(g) \rightarrow \text{HOBr}(g)$
 ii. $\text{HOBr}(g) + \text{HBr}(g) \rightarrow 2 \text{H}_2\text{O}(g)$
 iii. $\text{HOBr}(g) + \text{HBr}(g) \rightarrow \text{H}_2\text{O}(g) + \text{Br}_2(g)$

- (a) Please show how the elementary reactions are combined to give the overall reaction.
 (b) Based on the experimentally determined rate law, which step is rate determining?
 (c) What are the intermediates in this mechanism?
 (d) If you are unable to detect HOBr or HOOBr among the products, does this disprove the mechanism?

C3. Consider the simple salts NaCl(s) and AgCl(s). We will examine the equilibria in which these salts dissolve in water to form aqueous solutions of ions:



Substance	Cl ⁻ (aq)	Na ⁺ (aq)	Ag ⁺ (aq)	NaCl(s)	AgCl(s)
ΔG_f° (kJ/mol)	-131.2	-261.9	77.11	-384.0	-109.7
ΔH_f° (kJ/mol)	-167.2	-240.1	105.90	-410.9	-127.0
S° (J/mol-K)	56.5	59.0	73.93	72.33	96.11

- (a) Calculate the value of ΔG° at 298 K for each of the preceding reactions.
 (b) The two values from part (a) are very different. Is this difference primarily due to the enthalpy term or the entropy term of the standard free-energy change?
 (c) Use the values of ΔG° to calculate the K_{sp} values for the two salts at 298 K.
 (d) How will ΔG° for the solution process of these salts change with increasing temperature? What effect should this change have on the solubility of the salts?

($R = 8.314 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$)

Part B: Short Answer – Write legibly in each of the following problems. (24 Pts)

Number of questions: 6

Points per question: 4

B1. (a) Give the conjugate base of the following Brønsted–Lowry acids: (i) H_2PO_4^- , (ii) HI .
(b) Give the conjugate acid of the following Brønsted–Lowry bases: (i) CN^- , (ii) H_2PO_4^- .

B2. Indicate the principal type of solute–solvent interaction in each of the following solutions and rank the solutions from weakest to strongest solute–solvent interaction.

(a) KCl in water

(b) CCl_4 in cyclohexane (C_6H_{12})

(c) methanol (CH_3OH) in water

B3. For the elementary process $\text{N}_2\text{O}_5(\text{g}) \rightarrow \text{NO}_2(\text{g}) + \text{NO}_3(\text{g})$, the activation energy (E_a) and overall ΔE are 154 kJ and 136 kJ, respectively.

(a) Draw the energy profile for this reaction, and label E_a and ΔE .

(b) What is the activation energy for the reverse reaction?

B4. At a temperature of 700 K, the forward and reverse rate constants for the reaction $2 \text{HI}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{I}_2(\text{g})$ are $k_f = 1.8 \times 10^{-3} \text{ M}^{-1}\text{s}^{-1}$ and $k_r = 0.063 \text{ M}^{-1}\text{s}^{-1}$.

In this case, both the forward reaction and the reverse reaction are elementary reactions.

What is the value of the equilibrium constant K_c at 700 K?

Is the forward reaction endothermic or exothermic if the rate constants for the same reaction have values of $k_f = 0.097 \text{ M}^{-1}\text{s}^{-1}$ and $k_r = 2.6 \text{ M}^{-1}\text{s}^{-1}$ at 800 K? Explain.

B5. How do following changes affect the equilibrium constant K_p for a gas-phase exothermic reaction.

(a) increase in the total pressure by decreasing volume

(b) removal of a product

B6. You are asked to prepare a pH = 4.80 buffer, and you have the following 0.10 M solutions available: HCOOH , CH_3COOH , H_3PO_4 , HCOONa , CH_3COONa , and NaH_2PO_4 . Which solutions would you use? Explain.

Formula	Dissociation Constants K_{a1}
HCOOH	1.8×10^{-4}
CH_3COOH	1.8×10^{-5}
H_3PO_4	7.5×10^{-3}